



David Vilela

Junior & Entry-level Software Developer

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🌐 <https://davidvilela-dev.vercel.app>

Summary

Software Developer focused on TypeScript, React, Node.js and backend systems. Self-taught developer with a strong interest in software architecture, developer tooling and systems programming. I enjoy building software from first principles and exploring how systems work internally.

Languages

English

Fluent



Portuguese

Native



Technical Skills

Programming Languages

Typescript, Javascript, Python, Rust, SQL



Frontend

React, Next.js, Tailwind CSS



Backend

Node.js, REST API's, FastAPI



Databases

PostgreSQL, SQLite, MongoDB



Developer Tools

Git, GitHub, VS Code Extension Development, TypeScript Compiler API



Engineering

Software Architecture, API Design, Automated Testing, Concurrency / Asynchronous Programming



Interests

Electronics & Hardware

Computers, Robotics, Arduino, Electronic components

Retro-gaming

PS2, PS1, XBOX

Content Creation

Projects

Hound – Network Monitoring & Analysis Platform

Python · FastAPI · WebSockets · SQLite · Scapy

Built a local network monitoring platform for DNS/TCP traffic analysis, event processing and risk assessment. Designed a concurrent ingestion pipeline using bounded queues and dedicated processing workers, with a WebSocket-based dashboard for real-time events.

- Designed an event-driven architecture separating packet capture, processing and API layers.
- Implemented concurrent processing and persistence with explicit backpressure handling.
- Added automated tests, CI and technical documentation.

<https://github.com/DavidVilela1/hound>

TSBlueprint – TypeScript Code Visualization Tool

TypeScript · TypeScript Compiler API · VS Code API · WebView

Built a VS Code extension that parses TypeScript source code through the TypeScript Compiler API and generates interactive visual representations of types and interfaces.

- Built AST-based analysis instead of regex-based source parsing.
- Designed communication boundaries between the VS Code extension host and WebView.
- Implemented validation, debouncing and automated tests.

<https://marketplace.visualstudio.com/items?itemName=davidtools.ts-blueprint>

PurplePie – 2D Game Engine

Rust · wgpu · ECS · winit

(Work in Progress)

Developing a modular 2D game engine in Rust, exploring ECS architecture, rendering, input handling, fixed-timestep simulation and engine/game separation.

- Designed a modular architecture around ECS and explicit engine stages.
- Documented architectural decisions, risks and implementation roadmap.

Video Essay, Video Editing, Voice Acting

Cars & Mechanics

JDM, GDM, Classics

Inde Game Development

Godot, Unity, Unreal

- Exploring low-level systems programming and graphics architecture in Rust.

<https://github.com/DavidVilela1/purplepie>

Experience

Freelancing

Worldwide

Freelance Programmer

2024-Present

- Develop custom web applications and software solutions for clients.
- Work across frontend and backend development, from requirements and implementation to delivery.
- Build and maintain applications using modern web technologies.
- Communicate directly with clients to understand requirements and translate them into technical solutions.

<https://www.upwork.com/freelancers/~01d2a5fc7c326f3193>

Education

ISLA Gaia

CTeSP

Web & Mobile Development

Vila Nova de Gaia • 2024-2026

Program not completed